

BANKING TRANSCATIONS FRAUD DETECTION

**USING SQL
& EXCEL**

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- **Role : Data Analyst (Fresher Project)**
- **Tools Used : SQL server, microsoft Excel**
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INTRODUCTION :

Banking institutions handle a large number of transactions every day. Some of these transactions may be **fraudulent** due to unauthorized access, card cloning , or misuse.

This project focuses on identifying suspicious banking transactions using:

- SQL for data extraction and rule application
- Excel for analysis and summarization

The goal is to detect **potential fraud patterns** using a rule-based approach.

Problem statement

- Fraudulent transactions can cause:
 1. financial loss
 2. Customer dissatisfaction
 3. security risk

Problem :

How can we identify suspicious transactions using available transaction data ?

Solution:

Apply business-based fraud rules using SQL and excel to flag risky transactions and accounts.

DATASET DESCRIPTION

- The dataset contains banking transaction details.

Table Name : `bank_transactions`

Columns Description:

- `Transaction_ID` - unique transaction number
- `Account_Id` - customer account identifier
- `Transaction_Date` - Date of transaction
- `Transaction_Time` - Time of transaction
- `Amount` – Transaction amount
- `Transaction_type` - Debit/ credit
- `Location` – City of transaction
- `Channel` - ATM / Mobile/ Web
- `Status` – Success /Failed

Each row represents **One Transaction.**

TOOLS & TECHNOLOGIES

- SQL Server (ssms)

Used to create tables and apply fraud detection queries.

- Microsoft Excel

Filters

IF formulas

Pivot Tables

Summary analysis

These tools are commonly used by **data analysts in banking domains**



FRAUD DETECTION APPROACH

1. A rule-based fraud detection approach was used.

Steps followed:

1. Extract transaction data using SQL
2. Export data to Excel
3. Apply fraud rules
4. Identify suspicious transactions/accounts
5. Summarize insights for reporting

Fraud rule 1 : HIGH Amount transactions

- **Business logic:**
 - Very high transaction amounts may
 - indicate fraud.
- **Rule Applied:**
 - Transaction with **Amount > 10000**
 - Method used:
 - SQL where condition
 - Excel filter
- **Outcome:**
 - High- value risky transactions were identified.

FRAUD RULE-2 : ODD HOUR TRANSACTIONS

- Business logic:
- Fraud often occurs during late night hours.
- Rule Applied:
- Transactions between 12:00 A.M and 5:00 A.M
- Method used:
- SQL count,group by and having funtions
- Excel IF and TIME function
- Outcome:
- Night –time suspicious transactions were flagged.



FRAUD RULE 3: FAILED TRANSACTIONS

- Business logic:
- Repeated failed transactions may indicate PIN guessing or unauthorized access.
- Rule Applied:
- Accounts with more than one failed transaction
- Method used :
- Excel pivot table
- Count of failed transactions
- Outcome:
- Accounts with repeated failures were analyzed.



FRAUD RULE 4 : MULTIPLE LOCATIONS

- Business logic:
- Same account used in multiple locations may indicate card cloning.
- Rule applied:
- Accounts used in more than one location
- Method Used:
 - Excel pivot table
 - Distinct count of locations
 - IF Condition (count >1)
- Outcome :
- Suspicious multi – location accounts were identified

FRAUD SUMMARY (EXCEL INSIGHTS)



This summary helps decision-makers quickly identify fraud risks.

FRAUD RULE	RESULT	RISK IDENTIFIED
High amount	Yes	High-value misuse
Odd hour	Yes	Late –night activity
Failed Transactions	Limited	PIN misuse
Multiple Locations	Yes	Card cloning

BUSINESS IMPACT



EARLY FRAUD
DETECTION



REDUCED FINANCIAL
RISK



IMPROVED CUSTOMER
SECURITY

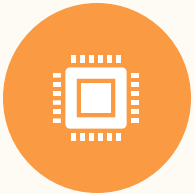


SUPPORTS FRAUD
MONITORING TEAMS



RULE -BASED FRAUD
DETECTION IS SIMPLE
YET EFFECTIVE.

CONCLUSION



This project demonstrates how SQL and Excel can be used to detect fraudulent banking transactions.



By applying business-driven rules, suspicious transactions and risky accounts were successfully identified.



This approach is widely used in real-world banking environments.